Hw #5 AUSTIN SILAGHI

a) $\{w\#t \mid w$ is a substring of $t\}$

Assume contrary. By Pumping lemma

$|s| \geq p$ $\quad s = uvwxy$

- $|vwx| \leq p$ $\quad$ - $|vx| \geq 1$. uv^nxy^nz $\quad n \geq 0$

$s = a^p \# a^{2p}$

$s = uvxyz$ $\quad |vwx| \leq p$

since v and y are only part of w or t not both. The pumping disrupts the substring relationship

- $p = 3$, so $s = aaa\#aaaaaa$

$s = uvxyz$, $n = 2$

$uv^2xy^2z = aaaa\#aaaaaa$ $\quad w \neq t$.

b) $\{a^n\#a^{2n}\#a^{3n} \mid n \geq 0\}$

$s = uv^nxy^nz$

v and y are in first segment a^n.

Pumping only increases a^n; $a^{2p}\#a^{3p}$ remains the same

new pumped string doesn't hold same ratio $a^n : a^{2n} : a^{3n}$,

$s \notin L$

Ex. $p = 3$ $\quad s = aaa\#aaaaaa\#aaaaaaaaa$

$s = uv^2xy^2z = aaaa\#aa...\#a...$

not $1:2:3$ ratio.